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Sexual Dysfunction and Disability

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SUMMARY PEARLS

- The *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition, classifies sexual dysfunctions based on criteria requiring significant distress and a minimum duration of 6 months, emphasizing that sexual issues in patients with disabilities only need treatment if they negatively impact the patient's quality of life.
- Sexual dysfunction in disability and chronic disease stems from various factors: primary physical changes, secondary physical limitations, psychosocial issues, comorbid conditions, and medication side effects.
- Sexual dysfunction in spinal cord injury varies based on injury level and completeness.
- Sexual dysfunction after a stroke is common in both males and females, with males often experiencing erectile and ejaculatory dysfunction, and females facing decreased sexual drive and vaginal lubrication.
- Sexual dysfunction after traumatic brain injury includes reduced sexual desire and activity, erectile dysfunction in males, and dyspareunia and anorgasmia in females.
- Conditions such as multiple sclerosis and Parkinson disease can lead to sexual dysfunction, affecting both males and females. Peripheral neuropathy, often seen in conditions such as diabetes and Guillain-Barré syndrome, can also cause sexual dysfunction.
- Cancer treatments, including chemotherapy, hormonal therapies, radiation, and surgery, often lead to sexual dysfunction.
- Chronic pain conditions such as rheumatoid arthritis and osteoarthritis can impact sexual function through joint pain, stiffness, fatigue, and medication effects.
- Limb loss can present sexual challenges related to factors such as depression, poor self-esteem, phantom sensations, and difficulties with body positioning during sexual activity.

Introduction

Sexuality is one of the most complex aspects of human life, with myriad physiologic and psychosocial influences. It is a topic that may be challenging to discuss comfortably in a physician's office. It may promote feelings of embarrassment and discomfort on the part of the patient and the physician, and yet it is one of the most important determinants of quality of life. Disability often has a dramatic negative impact on sexual functioning, but it does not change the inherent human yearning for intimacy, companionship, and physical satisfaction that sexuality affords. The physician who has the willingness and knowledge to address issues of sexual dysfunction openly with patients will enrich their lives immeasurably. The goal of this chapter is to provide an overview of sexual function and dysfunction, with an emphasis on the sexual challenges faced by people with disabilities. Great care has been taken to give equal treatment to the sexual dysfunctions of both males and females because female sexuality is often overlooked both in society and in the medical literature.

Sexual Response and Behavior

Human Sexual Response

The classic model of human sexual response was formulated by Masters and Johnson¹⁶¹ in the 1960s on the basis of a study of

600 able-bodied males and females. The model depicts both sexes as having similar sexual responses throughout four phases: excitement, plateau, orgasm, and resolution (Box 23.1).¹⁶² In the Masters and Johnson model, males tend to pass through each phase faster than females and achieve only one orgasm per cycle, often with a very short plateau phase. Females can achieve multiple orgasms in the same sexual response cycle. Masters and Johnson¹⁶¹ also noted that females could become "stalled" at the plateau phase and then pass straight to resolution without achieving orgasm.

There have been many critics of the Masters and Johnson sexual response model, primarily because it places too much emphasis on genital responses and does not acknowledge the role of central neurophysiologic control.^{17,206,257} In the late 1970s, Kaplan¹²⁹ devised a new model of sexual response with three phases: desire, excitement, and orgasm. In the Kaplan model, desire always precedes arousal and is described as "the specific sensations that motivate the individual to initiate or become responsive to sexual stimulation."^{129,141,206} The Kaplan model, similar to that of Masters and Johnson, proposes that human sexual response is linear and basically invariant between males and females.¹⁴¹

More recent research has furthered our understanding of human sexual response, particularly the ways in which the sexual response cycle is different between females and males. The Masters and Johnson and Kaplan models ignore major components of females' sexual satisfaction, such as the importance of trust,

• BOX 23.1 Masters and Johnson Model of Sexual Response

Excitement

- Associated with the senses, memory, and fantasy
- Increases in heart rate, blood pressure, and respiratory rate
- Myotonia develops late in the arousal phase
- **Males:** Engorgement of the corpus cavernosa of the penis, testicular elevation, scrotal skin flattening, and possible nipple erection
- **Females:** Clitoral enlargement, vaginal lubrication, constriction of the lower third of the vagina with dilation of the upper two-thirds, uterine elevation out of the deep pelvis, nipple erection, and areolar enlargement

Plateau

- Further increases in heart rate, respiratory rate, and muscle tone
- Sex flush may develop (rash over the chest, neck, and face)
- **Males:** Increase in diameter and color of the glans penis and increase in testicular elevation and size of 50% to 100% over baseline
- **Female:** Breast engorgement by up to 50% and clitoral shaft and glans retraction
- Lasts seconds to minutes and described as a “general sense of well-being”

Orgasm

- Further increases in heart rate, respiratory rate, and blood pressure
- Involuntary rhythmic contractions of the pelvic floor muscles
- **Males:** Only one orgasm per cycle
- **Females:** Can have multiple orgasms per cycle or can skip this phase altogether and go straight from plateau to resolution

Resolution

- Generalized perspiration in conjunction with gradual reversal of the previously mentioned changes in heart rate, blood pressure, and respiratory rate
- Lasts 5 to 15 minutes, but the penis does not return to its normal size for 30 to 60 minutes after orgasm

From Masters WH, Johnson VE. *Human Sexual Response*. Bantam Books; 1980.

intimacy, affection, respect, and communication.¹⁷ Basson^{17–19} proposed a new model for the female sexual response to address these sex differences (Fig. 23.1).¹⁴¹

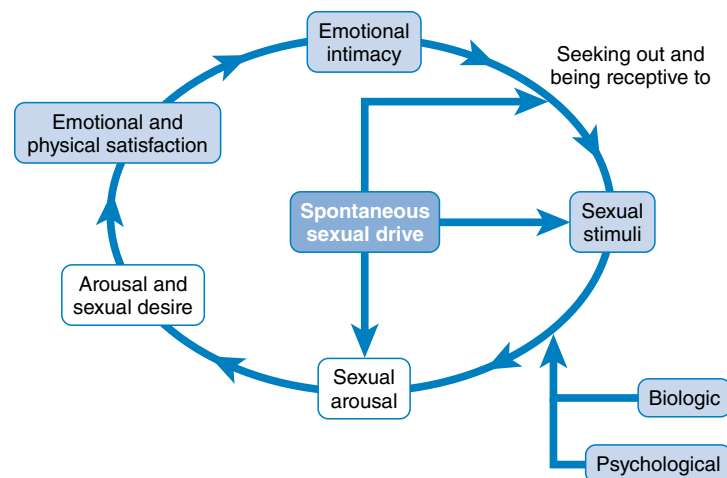
Basson's circular model emphasizes that sexual response in females is much more complex than that in males. In females, a sexual encounter does not necessarily start from a place of spontaneous

sexual drive or desire. Females often approach becoming intimate from a point of sexual neutrality, and the decision to become sexually engaged can result from numerous and varied factors, including a wish to connect emotionally with their partner.^{17,141} Sexual arousal and sexual satisfaction often do not occur solely through physical means, such as clitoral stimulation and orgasm, but may also be dependent on intangible factors, such as the ability to focus the mind on the present moment and a feeling of security or psychological well-being.²¹ The circular sexual response cycle may be repeated many times within the same sexual encounter.²¹ Janssen et al.¹²¹ have also proposed a circular model of human sexual response that is similar to that of Basson, which might be more applicable to both sexes.¹²⁷

Sexual Behavior and Aging

The frequency of sexual activity has been well documented to decline with age.^{71,126} Recent research has shown that the degree of decline is much less than was previously thought and that sexuality remains an important contributor to quality of life throughout the entire lifespan.^{71,126} An important reason for the discrepancy is that older studies tended to quantify sexual activity only as intercourse, but more modern research looks at all aspects of sexuality.^{71,126} The 2009 National Survey of Sexual Health and Behavior found that although sexual activity declined with age, in participants age 70 years or older, 43% of males and 22% of females reported vaginal intercourse in the year before the study.¹¹² Solo masturbation was reported in 46% of males and 33% of females in this age group.¹¹²

Many medical factors influence sexual activity in the elderly, including sexual dysfunction caused by medical illness, increasing frailty, and the side effects of medications. Postmenopausal females tend to experience vulvovaginal atrophy and vaginal dryness, which cause pain with vaginal penetration.¹²⁷ Males have decreased testosterone with advancing age, which contributes to diminished sexual drive and physical effects that contribute to increased frailty.¹²⁶ Psychosocial barriers to sexual activity in the elderly are numerous, including decreased partner availability, alterations in body image and change in self-perception, cognitive decline, caregiver burden, and environmental issues, such as the loss of privacy experienced in many residential settings.^{86,126} Age-related sensory dysfunction has been shown to be associated



• **Fig. 23.1** Female sexual response cycle. (Redrawn from Kingsberg SA, Janata JW. Female sexual disorders: assessment, diagnosis, and treatment. *Urol Clin North Am*. 2007;34:497-506.)

with sexual inactivity but not with sexual motivation or sexual expression.²⁷³

Aging patients in the lesbian, gay, bisexual, transgender, and queer (LGBTQ) population face unique challenges in regard to health and sexuality, including increased prevalence of smoking, poorer mental health, and functional limitations.^{81,89} These disparities are not experienced uniformly: LGBTQ subgroups, including adults older than age 80 years, transgender older adults, and those living with human immunodeficiency virus (HIV), are disproportionately affected. Health care providers must be aware of these disparities and be familiar with basic terminology regarding sexual orientation and expression. It is recommended to clarify a patient's preferred pronouns rather than relying on appearance or what the medical record may indicate.¹²²

Types of Sexual Dysfunction

Sexual dysfunction is most frequently classified according to the *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition (DSM-5), which was published in 2013.¹⁰ To qualify as a sexual dysfunction, a person's sexual problem must cause "significant distress."¹⁰ This distinction is important to remember for patients with disabilities because they might experience altered sexual response but not have a sexual dysfunction in need of further workup or treatment.²³⁶ In the same way that spasticity after stroke or phantom sensations after amputation do not always need to be addressed by the physician, a patient's sexual difficulties only need to be treated when the patient's quality of life is adversely affected.

The DSM-5 stipulates that sexual dysfunction diagnoses require a minimum duration of 6 months and that symptoms must be present 75% to 100% of the time (with the exception of medication-induced sexual dysfunction disorders). The DSM-5 also allows for two subtypes to be applied to all primary diagnoses to further clarify the nature of the dysfunction. The first subtype describes the onset of the disorder: lifelong or acquired (meaning it developed after a period of normal functioning). The second subtype is used to designate the context in which the dysfunction occurs: generalized or situational (meaning limited to certain types of stimulation, situations, or partners). The patient's sexual dysfunction should not be able to be better explained by a "non-sexual mental disorder, a consequence of severe relationship distress (e.g., partner violence), or other significant stressors."¹⁰ The DSM-5 criteria highlight the importance of separate classification systems for males and females and abandons the Masters and

Johnson linear sexual response model for both sexes.²⁵² The DSM-5 sexual dysfunctions are summarized in Table 23.1.¹⁰ In addition to the DSM-5, there have been a number of international consensus conferences that have helped to further define and classify sexual dysfunctions.^{22,23,26,155,178}

Male Sexual Dysfunction

The focus of both clinical care and research for males with sexual dysfunction has traditionally centered on performance problems, particularly erectile dysfunction/disorder (ED) and premature ejaculation. There has been a major paradigm shift in the past 2 decades away from the psychiatric understanding of most sexual disorders in males and toward a medicalization of male sexuality (particularly with the advent of proerection medications).^{165,209} Purely psychogenic male sexual disorders are much more underrepresented in the medical literature and are often misdiagnosed as ED by health physicians.¹⁷¹ Types of male sexual dysfunction include:

- **Male hypoactive sexual desire disorder:** Persistent or recurrent absence or deficit of sexual fantasies and desire for sexual activity, accounting for factors that affect sexual function, such as age and life context.¹⁰ Hypoactive sexual desire disorder has been reported in 0% to 15% of males, with a metaanalysis by Simons and Carey²³³ indicating a prevalence of 0% to 3% from community samples.^{171,206,233}
- **Erectile disorder/dysfunction:** Persistent or recurrent inability to attain or maintain until completion of the sexual activity, an adequate erection.¹⁰ The diagnosis is primarily based on the patient's self-report, and ED must occur on a recurrent basis to establish the diagnosis (unless there is a history of trauma or surgically induced ED).^{10,178} Prevalence estimates for ED vary greatly (0–5% to 52%).^{80,224,233} Results of the Massachusetts Male Aging Study in 1994 showed that 35% of males age 40 to 70 years demonstrated moderate to severe ED, with another 25% exhibiting mild symptoms.⁸⁵ The 2006 National Health and Nutrition Examination Study was tailored to create a racially diverse sample population and found ED in one of every five respondents older than age 20 years.²¹⁶ ED is known to be associated with advanced age, cardiovascular disease, diabetes, dyslipidemia, smoking, obesity, and depression; therefore the prevalence of this disorder is predicted to grow in keeping with the prevalence of these conditions.^{173,196,224} ED was previously known as impotence, but the term has been phased out because of its pejorative implications and lack of precision.^{185,224}

TABLE
23.1

Sexual Dysfunctions in the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5)

Male Dysfunctions	Female Dysfunctions	Other Dysfunctions
Male hypoactive sexual desire disorder	Female sexual interest/arousal disorder ^a	Substance/medication-induced sexual dysfunction
Premature (early) ejaculation	Female orgasmic disorder	Other specified sexual dysfunction
Delayed ejaculation (formerly known as male orgasmic disorder)	Genitopelvic pain/penetration disorder	Unspecified sexual dysfunction
Erectile disorder		

^aDesire and arousal classifications merged in DSM-5,¹¹ as compared with DSM-IV-TR.¹⁰

From American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*. 5th ed. American Psychiatric Publishing, 2013.

- *Premature (early) ejaculation disorder*: Persistent or recurrent ejaculation before or within approximately 1 minute of intravaginal penetration or clinically significant reduction in latency time (about <3 minutes), inability to delay ejaculation in all vaginal encounters, and negative personal consequences.^{10,173} This contrasts the average intravaginal ejaculation latency time (IELT) of 5.4 minutes in multinational populations.²⁶² Factors that affect duration of the excitement phase must be taken into account, such as age, novelty of the sexual partner or situation, and recent frequency of sexual activity. Consensus opinions also recognize that psychological distress in the patient and/or in the patient's partner is a critical diagnostic component.^{6,178,210} Prevalence data for premature ejaculation are often limited to self-report, with estimates of up to 30% across all age cohorts.^{152,207,209} However, when the IELT time parameter is applied to diagnostic criteria, prevalence drops to 1% to 3%.⁶ Approximately 30% of males with premature ejaculation have concomitant ED, which typically results in early ejaculation without full erection.¹¹¹ Premature ejaculation is often caused by a combination of organic, psychogenic, and relationship-based factors.^{6,210} More recently, neurobiologic and genetic variations have been proposed as contributing factors in premature ejaculation.⁶
- *Delayed ejaculation*: Persistent or recurrent unwanted delay in, or absence of, ejaculation after a normal sexual excitement phase during sexual activity, which is adequate in focus, intensity, and duration.^{10,169,178} This condition was formerly termed male orgasmic disorder.⁹ Delayed ejaculation can originate through a variety of anatomic, neurogenic, and endocrine-related causes, as well as medication-related side effects and psychogenic etiologies.¹⁶⁹ As the least common sexual complaint among males, the prevalence of delayed ejaculation in the general population is thought to be in the range of 0% to 9% and is more likely to affect partnered vs. masturbatory ejaculatory responses.^{169,233}
- *Dyspareunia*: Recurrent or persistent genital pain associated with sexual intercourse. Male dyspareunia was eliminated as a separate entity in the DSM-5 (compared with previous editions of the DSM).^{9,10,252} This change was apparently made because dyspareunia in males is much less prevalent than in females; however, estimates do range from 0.2% to 8%.^{233,252} In males who are gay, the prevalence has been estimated at 3% (insertive) and 16% (receptive).²⁰⁸

Female Sexual Dysfunction

Female sexual dysfunction is very common, with a reported prevalence of 40% to 50% in multiple population-based studies.^{12,152} Sexual dysfunction in females has become a focus of renewed research interest in the past 10 to 15 years in part based on the new understanding of the female sexual response cycle proposed by Basson¹⁷ and the implications of that understanding toward diagnosis and treatment.^{17,141} In contrast to sexual dysfunction in males, the psychological aspect of females' sexual functioning typically receives far more attention than organic etiologies of dysfunction. This discrepancy is partially because of institutional bias, but it is also based on a growing body of data suggesting that psychological factors correlate more strongly with sexual dysfunction in females than do medical problems.^{26,141} It is especially important to remember that personal distress is necessary to make a diagnosis of sexual dysfunction when dealing with female patients. Up to 50% of females who report a problem with

sexual functioning do not have any associated personal distress, and thus they cannot be classified as having a sexual dysfunction.²²⁵ Types of female sexual dysfunction include:

- *Female sexual interest/arousal disorder*: With updated DSM-5 criteria, female sexual interest/arousal disorder is a convergence of the female desire and arousal disorders formerly termed female hypoactive sexual desire disorder and female sexual arousal disorder in the revised fourth edition of the DSM (DSM-IV-TR).^{9,10} For diagnosis, at least three of the following six criteria must be absent or reduced: (1) interest in sexual activity, (2) sexual/erotic thoughts or fantasies, (3) initiation of sexual activity or receptiveness to a partner's initiation attempts, (4) sexual excitement/pleasure during sexual encounters, (5) sexual interest/arousal response to internal/external sexual/erotic cues, and (6) genital/nongenital sexual activity sensations during almost all or all sexual encounters.¹⁰ Owing to the recent classification of female sexual interest/arousal disorder as a unique disorder, the prevalence is currently unknown, and there are limited research/data on interventions and treatment.⁵⁷ Prevalence data for the previous DSM-IV-TR disorders of female hypoactive sexual desire disorder and female sexual arousal disorder have been well established:
 - Female hypoactive sexual desire disorder was previously defined as absent or diminished feelings of sexual interest or desire, absent sexual thoughts or fantasies, and a lack of responsive desire.⁹ Lack of interest was described as being beyond what would be considered normal for the female's age and relationship duration and must have been a frequent and persistent problem not related to changes in life situation or relationship dynamics.^{22,225} The lack of responsive desire was the key to diagnosis because many normal females do not have spontaneous sexual thinking, fantasizing, or desire ahead of sexual activity.²⁰ Despite traditional notions, female hypoactive sexual desire disorder has not been clearly linked to menopause or advancing age.^{109,141,225} Prevalence estimates in population-based studies have been typically quoted at 24% to 43% but may be much lower when strict diagnostic criteria are applied (closer to 5.4–13.6%).²²⁵ Hypoactive sexual desire disorder has been associated with lower health-related quality of life, lower general happiness and satisfaction with partners, and more frequent negative emotional states.¹⁹¹
 - Female sexual arousal disorder was previously defined as persistent or recurrent inability to attain or maintain until completion of the sexual activity and sufficient sexual excitement, which may be expressed as a lack of subjective excitement or as a lack of adequate genital (lubrication-swelling) or other somatic response.^{9,23} The prevalence has been estimated to be 6% to 21%.²³³
- *Female orgasmic disorder*: Persistent or recurrent lack of orgasm, markedly diminished intensity of orgasmic sensations, or marked delay of orgasm from any type of stimulation despite the self-report of high sexual arousal/excitement.^{10,26} Orgasmic disorder in females is often difficult to distinguish from an inadequate arousal response without a careful history because females with decreased arousal will also not be able to achieve orgasm.¹⁴¹ Many females are situationally orgasmic, in that they can reliably achieve orgasm with some forms of stimulation but not others. Vaginal intercourse alone is not a reliable way for many females to achieve orgasm, and the need for clitoral stimulation to achieve orgasm is not considered abnormal.¹⁴¹ The prevalence of female orgasmic disorder was

estimated at 24% in the National Health and Social Life Survey.¹⁵² Other estimates range from 4% to 42% based on research setting and methodology.^{202,233}

- **Genitopelvic pain/penetration disorder:** Defined by persistent or recurrent difficulties with vaginal penetration, genitopelvic pain during vaginal intercourse or other penetration attempts, fear or anxiety of genitopelvic pain or vaginal penetration, and/or increased pelvic floor tension during attempted vaginal penetration.¹⁰ This disorder encompasses the former DSM-IV-TR disorders of dyspareunia (persistent or recurrent pain with attempted or complete vaginal entry and/or penile vaginal intercourse) and vaginismus (the persistent or recurrent difficulties of the female to allow vaginal entry of a penis, a finger, and/or any object, despite the expressed wish to do so).^{9,22} Vaginismus was originally defined as persistent spasm of the outer third of the vaginal musculature, which interferes with vaginal penetration. More recent research and consensus opinion dispute the classic definition because muscle spasm and even pain are not always present.^{22,141,201} Prevalence rates for vaginismus have been reported at 1% to 6%.¹⁵⁵ Vaginismus is one of many potential causes of dyspareunia. Dyspareunia was previously thought to be primarily of psychogenic origin, but it is now known that biologic factors often contribute to the presentation.¹⁴¹ Some even characterize dyspareunia as a pain disorder that interferes with sexuality instead of a sexual disorder characterized by pain.^{34,141} The differential diagnosis of dyspareunia in females includes vulvovaginal atrophy, inadequate lubrication, vaginitis, postmenopausal vaginal dryness, pelvic floor myofascial pain or dysfunction, vulvodynia (vulvar pain without visible signs of pathology and with no known etiologic diagnosis), and vestibulodynia/vulvar vestibulitis syndrome (a type of vulvodynia localized only to the vulvar vestibule, which may or may not have associated inflammatory changes visible on physical examination).^{25,100,114,141} Urethral disorders, cystitis, bladder pain syndrome (formerly known as chronic interstitial cystitis), anatomic variations, pelvic adhesions, infections, endometriosis, inflammatory bowel disease, abdominal wall pain, cancer, late effects of pelvic radiation, vulvovaginal dermatologic diseases (such as lichen sclerosis or lichen planus), and pelvic congestion syndrome can also cause chronic dyspareunia.^{25,141,264} Dyspareunia prevalence estimates range from 3% to 18% in the general population, 3% to 46% in a primary care setting, and 9% to 21% in postmenopausal females.²³³

Sexual Dysfunction in Disability and Chronic Disease

The development of sexual dysfunction after the onset of disability is usually related to myriad diverse etiologic factors that can be grouped into five different categories: primary physical changes, secondary physical limitations, psychosocial contributions, the effect of comorbid conditions, and medication-related factors. Primary physical changes derive from the disease pathophysiology and directly affect sexual response. These primary physical changes could include altered genital sensation, decreased vaginal lubrication, ED, or anorgasmia. Secondary physical limitations also play a significant role in the development of sexual difficulties after disability. These are indirect effects of the disease process that do not affect genital function or sexual response but which make sexual activity unpleasant or challenging. Examples of secondary

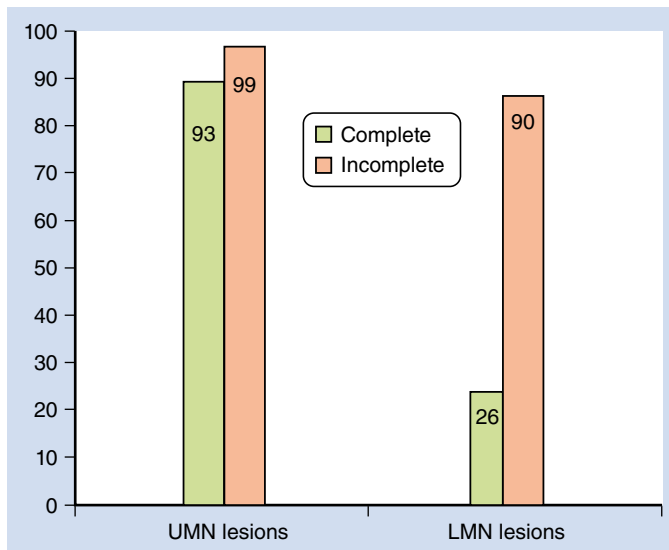
physical limitations include fatigue, pain, weakness, spasticity, contractures, ataxia, bowel and bladder dysfunction, and cognitive impairment. The psychosocial contribution to the development of sexual dysfunction includes psychological, emotional, relational, situational, social, and cultural factors. People with disabilities often have difficulty adjusting to their new life circumstances. They can have poor self-esteem and body image, fear of isolation and abandonment, anger, shame, guilt, strained relationships with loved ones who may also now be their caregivers, anxiety, and depression. All these issues can affect sexual functioning. In addition, the pathology of comorbid conditions such as diabetes, cardiovascular disease, and depression can affect sexual dysfunction independent of the primary pathology. Lastly, many medications commonly prescribed for people with disabilities can lead to the development of sexual dysfunction. It is important for the psychiatrist to consider potential etiologies from all five categories when diagnosing and treating a disabled patient presenting with sexual dysfunction. (Disability-specific sexual dysfunctions are discussed later.) The psychiatrist should also consider the impact of sexual dysfunction on the patient's partner. Partners may feel overwhelmed taking on a new role as caregiver for someone they are romantically involved with. They may experience challenges alternating between a caregiver and a sexual partner and may experience emotional distress with this. The physician should be mindful of this potential conflict when evaluating the patient for sexual dysfunction.

Spinal Cord Injury

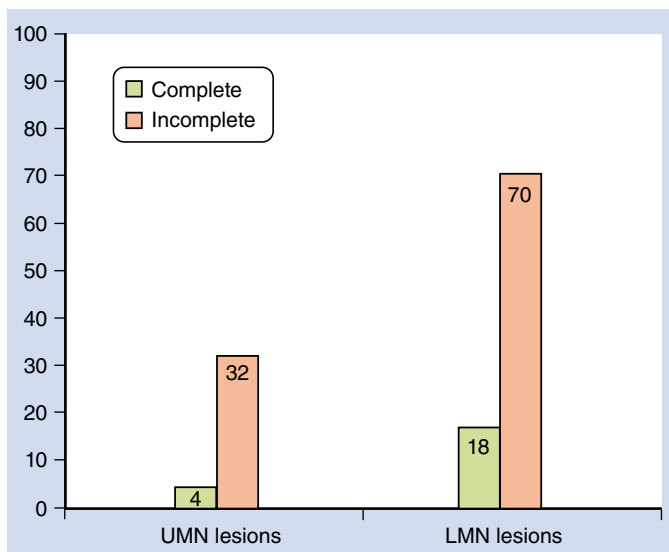
In contrast to many other disabilities, sexual dysfunction in patients with spinal cord injury (SCI) has been well studied in both sexes. The type of sexual dysfunction the patient will ultimately experience depends in large part on the spinal cord level and the degree of completeness of the injury.¹⁷⁷ Multiple studies have shown that the frequency of sexual activity and the level of sexual satisfaction decrease in both males and females after sustaining SCI.^{4,54,63,146,238} Sexual dysfunction after SCI may be caused by changes in desire or perceived desirability, concern regarding neurogenic bowel or bladder, genital arousal disorders, ejaculatory dysfunction, orgasmic disorders, and issues of pain.^{144,247} In studies of males with SCI, the main determinants of a positive sex life after injury have included the willingness to adapt their sexual life to injury and the experience of physical pleasure during foreplay and intercourse.²⁴³

Upper motor neuron (UMN) SCI allows for reflexogenic erection to occur in males because the parasympathetic sacral reflex arc is intact. Psychogenic erection, however, is usually not possible unless the injury is incomplete. Ejaculation is difficult for these patients because it is sympathetically mediated from T11 to L2. Lower motor neuron (LMN) SCI has preserved integrity of the thoracolumbar sympathetic outflow tract, and thus psychogenic erections and ejaculation are theoretically more likely to be intact, but reflexogenic erections are not.

Bors and Comarr⁴⁰ conducted a pivotal study in 1960 detailing the prevalence of ED and orgasmic dysfunction in 529 males with SCI. They found that in males with complete UMN lesions, 93% of patients were able to achieve reflexogenic erections, and none was able to attain psychogenic erections. Anterograde ejaculation in complete UMN SCI was seen in 4% of patients. Males with incomplete UMN SCI achieved erection 99% of the time, with 80% of males having only reflexogenic erections and 19% having combined reflex and psychogenic erections. Ejaculation



• **Fig. 23.2** Percentage of males with spinal cord injury able to achieve erection. *LMN*, Lower motor neuron; *UMN*, upper motor neuron. (Data from Bors E, Comarr AE. Neurological disturbances of sexual function with special reference to 529 patients with spinal cord injury. *Urol Surv.* 1960;110:191-221.)



• **Fig. 23.3** Percentage of males with spinal cord injury able to achieve anterograde ejaculation. *LMN*, Lower motor neuron; *UMN*, upper motor neuron. (Data from Bors E, Comarr AE. Neurological disturbances of sexual function with special reference to 529 patients with spinal cord injury. *Urol Surv.* 1960;110:191-221.)

was possible in 32% of patients with incomplete UMN SCI, with 72% of those occurring after reflexogenic erection and 26% occurring after psychogenic erection. Complete LMN lesions showed a very different picture, with 26% of patients able to achieve psychogenic erections, and none attaining reflexogenic ones. Ejaculation was possible in 18% of these males. Incomplete LMN lesions fared far better, with 90% retaining the ability to achieve erection and 70% the ability to ejaculate. The results of the Bors and Comarr study are summarized in Figs. 23.2 and 23.3.

Orgasmic ability has been shown to be preserved in 38% to 50% of males with complete UMN SCI, 78% to 84% of males with incomplete UMN injury, and 0% of males with complete LMN injury.^{4,5,194,235} It should be noted, however, that a majority of studies on males have focused on the ability to ejaculate instead of the ability to have an orgasm, which have more recently been noted to be separate occurrences.⁵ Some males are also occasionally able to achieve orgasm without anterograde ejaculation, possibly indicating either anejaculation or retrograde ejaculation into the bladder.²³⁵

Females with SCI have been shown to have similar sexual responses to males with arousal and vaginal lubrication as the female correlates to penile erection.^{63,239,240} In females with complete UMN injuries, reflexogenic, but not psychogenic, lubrication is preserved.^{63,240} Sensation to light touch and pinprick in the T11 to L2 dermatomes were found to be predictive of the ability for psychogenic vaginal lubrication, which is thought to be sympathetically mediated.^{63,239} From 44% to 54% of females with SCI have been shown to be able to achieve orgasm, although orgasm is much less likely in females with LMN injuries affecting the S3 to S5 segments.^{54,63,120,238,239} Self-exploration should be encouraged because studies suggest females with SCI may be able to achieve orgasm through stimulation of the anterior vaginal wall, cervix, and/or nipples.^{61,200,267}

Fertility in males with SCI is impaired in part because of a decreased ability to ejaculate.^{151,200} Semen quality has also been found to be poor in males with SCI, with decreased sperm motility, decreased mitochondrial activity, and increased sperm DNA fragmentation.²⁰³ Reasons for altered semen quality have been postulated to include seminal fluid stasis, testicular hyperthermia, recurrent genitourinary tract infections, and hormonal dysfunction.^{151,203} After initial postinjury decline, semen quality appears to plateau in the chronic postinjury period.^{41,118} Fertility in females with SCI is preserved once menstruation resumes, on average about 5 months postinjury.^{13,251} Family planning counseling should include these post-SCI fertility considerations.

Stroke

Stroke is the world's fifth leading cause of death and the leading cause of adult disability.²⁵¹ Sexual dysfunction after stroke is a common finding in both sexes. The most common findings for males are ED and ejaculatory dysfunction (in 40–50%) and decreased sexual drive.^{52,124,143,159,254} Women tend to experience decreased sexual drive, decreased vaginal lubrication (in approximately 50%), decreased orgasm (in approximately 20–30%), and decreased overall sexual satisfaction.^{52,138,143,159,254} Prevalence of decreased sexual drive has been reported to be between approximately 25% and 50% in both sexes after stroke, with similar effects in spouses.^{138,143} Stroke also leads to significantly decreased frequency of sexual activity in both males and females, and in one study, approximately 50% of stroke survivors reported no sexual activity whatsoever by 1 year poststroke.^{96,143,254}

The physical effects of stroke that contribute to sexual dysfunction include hemiparesis with its effect on body positioning and movement, hemineglect, hemianopsia, neurogenic bowel and bladder, spasticity, cognitive dysfunctions, communication constraints, and hygiene.^{179,182,200} Not much is known about the functional neuroanatomy of sexual behavior and control, but some studies have indicated that the right cerebral hemisphere may play a more important role in sexual functioning than the left side.^{60,94,125} Research suggests that multiple brain lesions are

associated with a higher prevalence of sexual dysfunction, specifically with ED.^{51,125} Psychological factors also contribute to the etiology of poststroke sexual dysfunction, including depression, anxiety, emotional lability, and changes in self-image, including perceived decline in sexual appeal.^{55,125,136,153,197} Many studies have suggested that sexual dysfunction after stroke is related to the presence of medical comorbidities (cardiac disease, diabetes, hypercholesterolemia, and depression), medications, and psychosocial factors (inability to discuss sexuality with the partner, unwillingness for sexual activity, fear of another stroke, and sexuality being unimportant) rather than the direct effect of stroke.¹⁸²

Traumatic Brain Injury

The prevalence of sexual dysfunction after traumatic brain injury (TBI) has been reported at 4% to 71%, this wide range likely attributed to the limited number of quality studies and the varied types and severities of injury.²¹⁸ In one TBI model systems multicenter study of post-TBI community-dwelling individuals, 20% met criteria for sexual dysfunction.²¹⁹ Greater impairments were correlated with female sex, older age, higher TBI severity, and decreased social participation.²¹⁹ Although sexual dysfunction prevalence and type may be associated with both global and focal brain tissue injury, previous studies have not shown a consistent relationship to injury severity and sexual dysfunction.^{39,200,218} Some researchers have postulated that hypopituitarism after TBI may have a larger contribution to sexual dysfunction than previously thought.^{105,123,167} Types of sexual dysfunction reported have been similar to those seen in patients with stroke, including reduced sexual desire and frequency of sexual activity in both sexes, ED and ejaculatory dysfunction in males, and dyspareunia, anorgasmia, and reduced lubrication in females.^{113,145,200,218,219} In addition, brain injury can cause hypersexual behavior, such as excessive masturbation. This can occur in particular with injury to the limbic system or prefrontal regions, which can cause disinhibition; or to bilateral temporal poles, which produces the Klüver-Bucy syndrome of hypersexuality and hyperorality.^{44,154,234} This hypersexual behavior can be treated with anticonvulsants, such as Depakote, but these medications can also greatly contribute to the sexual dysfunction witnessed after brain injury.⁹⁹ Psychosocial factors appear to play a major role in sexual functioning after brain injury with the presence of depression being the most sensitive indicator for sexual dysfunction.¹¹³ Other important psychological factors include perceived health status and quality of life, low self-esteem, anxiety, and perceived decline in personal sex appeal.^{113,147} Studies focusing on the partners and spouses of moderate to severe TBI survivors have also found disturbed sexual functioning for these individuals caused by a variety of psychosocial factors, including assuming a caregiver role, communication issues, and patient neuropsychiatric symptoms leading to a sense of “living with a stranger.”^{35,97,103}

Multiple Sclerosis

Multiple sclerosis (MS)–induced sexual dysfunction is present in 40% to 80% of females and 50% to 90% of males.⁷² Historically, sexual dysfunction has been attributed directly to the location and duration of the spinal cord or brain lesion, but more recent research tends to describe a multitude of other factors that also contribute to the development of sexual dysfunction in MS.^{45,150,245} Psychological and psychosocial causes have been proposed as primary drivers of sexual dysfunction in this population, although in

males an association has been found between insular lesions and ED.^{244,268} Research looking at health-related quality of life parameters in patients with MS demonstrated that sexual dysfunction had a more significant impact than severity of physical disability.²²¹

In MS, sexual dysfunction is now conceptualized into three levels of dysfunction.⁴⁵ Primary sexual dysfunction relates to neurologic impairments in libido, lubrication, and orgasm. Secondary sexual dysfunction includes the effects of fatigue and physical limitations on sexual function. Physical limitations may include bowel and bladder dysfunction, weakness, incoordination, positioning and body control during sexual encounters, numbness, paresthesias, pain, and cognitive impairment.^{150,245} Tertiary sexual dysfunction reflects the psychological, emotional, social, and cultural implications of MS on sexual function. Psychosocial factors contributing to tertiary sexual dysfunction include poor self-image, poor self-esteem, fear of isolation and abandonment, shame, dependency on the partner to provide for one's basic needs, and depression.^{222,245,261} In addition, medication side effects may contribute to sexual dysfunction.

Females with MS tend to have decreased sexual desire, anorgasmia, decreased vaginal lubrication, and increased spasticity with sexual activity.^{65,116,200} Many of these symptoms are thought to be related to decreased genital sensitivity experienced by 62% of females with advanced MS.¹¹⁶ Males with MS tend to have ED, ejaculatory dysfunction (premature, delayed, or absent), orgasmic dysfunction, decreased genital sensation, and decreased sexual drive.^{65,200} In one study, sexual dysfunction in males with MS was found to be correlated with lower-limb disability and bladder dysfunction, whereas in females, sexual disorders were most strongly correlated with fatigue.⁸⁸ Females report more symptoms of sexual dysfunction, whereas males report more dissatisfaction with sexual function.^{45,186}

Other Neurologic Disorders

Sexual dysfunction is a common but frequently overlooked non-motor symptom in Parkinson disease (PD) that significantly affects a patient's quality of life.^{139,220} Risk factors associated with developing sexual dysfunction include advanced age, male sex, and hormone deficiencies.¹⁸⁴ PD is often associated with low testosterone levels and therefore decreased sexual drive.¹⁷⁹ In one study of young males with PD (ages 36–56 years), up to 40% had low sexual desire.²⁶⁵ Males with PD also experience ED and premature or delayed ejaculation.^{46,200} In females, decreased lubrication, involuntary urination, decreased libido, and orgasmic dysfunction can occur during intercourse.^{46,184} In contrast, dopamine and dopamine agonist medications have been documented to cause hypersexual behavior, which often accompanies mania.²⁵⁹ Hypersexuality has also been described with deep brain stimulation of the subthalamic nucleus.²⁰⁴

Peripheral neuropathy can cause sexual dysfunction with most common etiologies, including diabetes, amyloid, and some inherited neuropathies with urogenital symptoms as prominent early clinical manifestations.²⁰⁰ Guillain-Barré syndrome has been linked to ED when there is residual neurologic deficit or disability after recovery.⁵⁰

Cancer

Owing to medical breakthroughs and technologic advancements in the diagnosis and treatment of cancer, there has been a significant increase in cancer survivorship in the United States. As of

2000, close to 10 million cancer survivors were alive in the United States, with approximately 62% of cancer survivors expected to live at least 5 years from their date of diagnosis.⁹² The majority of research on sexual dysfunction has been in regard to prostate and bladder cancer in males and breast and gynecologic cancers in females. However, patients with any cancer are susceptible to sexual side effects related to their cancer diagnosis or treatments.⁷⁶

Chemotherapy, hormonal treatments, radiation therapy, and surgical procedures can all affect sexual function in both sexes. Common chemotherapy agent side effects (fatigue, weakness, nausea, diarrhea) can contribute to decreased sexual desire. Additionally, chemotherapy and endocrine therapies have the capacity to induce ovarian failure and premature menopausal symptoms in females.⁷⁶ Removal of pelvic organs in females can cause irreversible menopause (in the case of oophorectomy), anatomic changes, dyspareunia, and a sense of loss regarding one's sexuality.²⁵⁰ Both lumpectomy and mastectomy for breast cancer have been associated with decreased positive sexuality owing to both physiologic and psychological sequelae.⁷⁶ In females, radiotherapy for breast cancer can cause significant skin changes and breast pain, whereas radiotherapy to the pelvis can lead to vaginal fibrosis resulting in subsequent pain and decreased sensation.²⁹ Androgen blockade is commonly used to treat prostate cancer but has been shown to contribute to both decreased libido and ED in males, with more than 80% of treated males reporting these symptoms 1 year after treatment.¹³⁰ Radiation therapy for prostate cancer can also cause ED.⁷⁶ Surgical prostatectomy has been associated with high rates of urinary incontinence and ED in males with additional possible long-term sequelae, including atrophy and fibrosis of the penile smooth muscle and degeneration of neural structures.^{31,130,223} Even after successful treatment of cancer, both males and females continue to report ongoing symptoms of sexual dysfunction, including decreased desire and arousal, ED, dyspareunia, and anorgasmia.^{11,77,110} Patients' sexual function may also be significantly impacted by changes in body image, fear of recurrence, and changed relationship dynamics.⁷⁶

Chronic Pain

The reasons for sexual dysfunction in patients with chronic pain are multifactorial, often related to physiologic, pharmacologic, and psychological factors.¹⁸⁹ In one study, 73% of patients with chronic pain had pain-related difficulty with sexual activity. The reasons for this included decreased arousal, positioning issues, exacerbations of pain, low confidence, performance worries, and relationship issues.⁸ In another study, sexual problems were reported in 46% of patients with low back pain.⁷⁹ Females with low back pain have been shown to have decreased frequency of sexual activity, more pain during sexual intercourse, and decreased sexual desire compared with males with low back pain or patients of either gender with neck pain.¹⁵⁷ Sexual dysfunction in patients with chronic pain has been correlated most strongly with the presence of depression, poor coping skills, and shorter pain duration.¹⁴⁸ Many of the medications used to treat chronic pain (including opioids, antidepressants, and anticonvulsants) can independently contribute to sexual dysfunction.²⁵⁵

Rheumatologic Disease

Osteoarthritis (OA) impacts sexual function primarily through associated joint pain, stiffness, and fatigue with the hip joint being the most commonly implicated in leading to sexual difficulties.^{36,64,179}

Total hip replacement often significantly improves sexual functioning; in one study, 65% of patients found relief from sexual difficulties after surgery.²⁴⁹ It is important to instruct patients in proper positioning after joint replacement to reduce the risk of dislocation, and many surgeons instruct patients to refrain from sexual intercourse for 1 to 2 months after hip replacement for this reason.²³⁷ Hip replacement and arthroscopy carry a risk of injury to the pudendal nerve with subsequent urinary and/or sexual dysfunction; however, the risk is low, and the vast majority of these cases resolve on their own.¹⁰⁶

Rheumatoid arthritis (RA), similar to OA, can cause problems in sexual functioning through joint pain, stiffness, and fatigue.¹⁸³ RA in males has also been shown to be associated with decreased sexual desire and ED during periods of active inflammation.¹⁰² One study of females with RA identified 62% with difficulties in sexual performance caused by joint pain and stiffness and 92% with diminished sexual desire or satisfaction. Difficulties in sexual performance were related to overall level of disability and hip involvement, whereas decreased sexual desire and satisfaction were correlated more with perceived pain, age, and depression.²

Amputation

Patients with limb amputation usually have preserved physiologic sexual functioning unless a comorbid condition (diabetes or cardiac disease) or medication side effects play a significant detrimental role.¹¹⁷ However, their sexual life can be significantly affected by a variety of factors associated with their amputation, including depression, poor self-esteem and body image, phantom sensations and pain, problems with balance and movement, and difficulties with body positioning during sexual activity.^{228,270} A study of males with lower limb amputation reported ED in nearly 70% of patients and orgasmic dysfunction in over 60%.³⁸ Preservation of the knee joint can be helpful for maintaining balance during sexual intercourse, although transfemoral amputees can use a pillow to aid in positioning.⁴⁸ Upper limb amputees would benefit from side-lying or supine positioning to allow for free movement of both the intact arm and the residual limb.²³⁷ Sexual dissatisfaction rates range from 13% to 75% among amputees with increased levels of sexual dysfunction despite preserved sexual desire. Therefore this topic should be integrated into postamputation reintegration counseling.^{38,93,117}

Diabetes Mellitus

Diabetes mellitus (DM) is one of the major comorbidities affecting people with disabilities because it is a significant risk factor for stroke and amputation. DM plays a clear role in the development of sexual dysfunction in males. ED is three times more common in males with DM than in the general population, with prevalence estimates ranging from 35% to 75%.^{27,158,245} Premature ejaculation and hypoactive sexual desire disorder have also been documented in 40% and 25%, respectively, of males with diabetes.¹⁵⁸ Diabetes-related sexual dysfunction in males is strongly correlated with glycemic control, duration of disease, and burden of diabetic complications.²⁷ The pathogenesis of sexual dysfunction in DM is likely to be multifactorial, including vasculopathy, autonomic neuropathy, and diminished nitric oxide production, which leads to decreased neurogenic vasodilation.^{12,245}

Diabetes in females has a more ambiguous effect on sexual functioning because research is limited and at times contradictory.¹⁷⁹ The most commonly reported condition is sexual arousal

disorder with decreased vaginal lubrication as a common symptom.¹² Low sexual desire, decreased clitoral sensation, orgasmic dysfunction, and dyspareunia have also been infrequently reported.⁵³ Factors for the development of sexual dysfunction in females with diabetes have been analyzed, and a far different picture has emerged than for males with diabetes. In females, there has been no documented correlation with diabetic complications, length of disability, or level of glycemic control.^{12,82,83} One meta-analysis demonstrated a relationship between decreased sexual function and elevated body mass index.¹⁹⁵ Associations have also been previously shown for cardiovascular comorbidity, depression, and age.²⁶³

Cardiac Disease

Cardiovascular disease is another common comorbidity in patients with disabilities. Hypertension, coronary artery disease, and congestive heart failure have all been associated with increased prevalence of sexual dysfunction.^{30,37,245} Hypertension is the most prevalent comorbidity among males with ED.^{49,245} ED has been noted in 40% to 95% of males with hypertension and is correlated with duration of hypertensive disease.^{85,174} Control of blood pressure often leads to restoration of erectile function.¹⁷⁹ ED is also common in cardiac disease with a prevalence of 42% to 75%.¹⁹⁸ ED is often one of the earliest indicators of cardiovascular disease, and males (with or without disabilities) who present with ED with no obvious cause should be worked up for this potentially deadly comorbidity.²⁴⁵ It has been reported that 25% to 63% of females with cardiac disease experience sexual dysfunction, including decreased sexual desire, vaginal dryness, dyspareunia, decreased genital sensation, and decreased orgasmic ability.^{3,30,245}

Psychological factors associated with cardiac disease also contribute to sexual dysfunction. Cardiac disease, especially after myocardial infarction, is often associated with depression and anxiety.^{90,101,205} Other psychological factors that are likely to play a role include fear of recurrence of cardiac symptoms or fear of death with resumption of sexual activity.¹¹¹ Cardiovascular deaths during sexual intercourse are actually very rare, with the rate in males estimated to be 0.2 per 100,000 and the risk in females 12 times lower than in males.¹⁹³

Many cardiovascular medications (including antihypertensives and digoxin) have been known to contribute to the sexual dysfunction seen in cardiac disease.²⁵⁵

Depression

Depression is one of the most common comorbidities in patients with all types of disabilities, and it has been shown to contribute to sexual dysfunction in a variety of ways, including medical, pharmacologic, and psychosocial.⁵⁶ There has been intense research and publicity about the sexual side effects of antidepressant medications, but untreated depression has also been strongly correlated with sexual dysfunction.²²⁶ Clinically, it can be difficult to distinguish whether a patient's sexual dysfunction is secondary to depression itself or to the antidepressant medication.¹⁵ In one study of 134 patients with major depressive disorder who were not taking any medication to treat their illness, hypoactive sexual desire was seen in 40% to 50% of males and females, decreased arousal was observed in 40% of females, ED was seen in 50% of males, and ejaculatory or orgasmic dysfunction was seen in 15% to 20% of males and females.¹³⁴ Psychosocial factors in patients with depression that are likely to play a role in the development

of sexual dysfunction include interpersonal relationship difficulties and poor body image and self-esteem.⁵⁶

Sexual Dysfunction Related to Medication Use in Individuals With Disability

Sexual dysfunction is a common side effect of many medications used routinely in patients with disabilities. It is important for the physician to maintain open communication with patients, understand these sexual side effects, and explain the risks to patients before prescribing these drugs. If sexual dysfunction occurs secondary to the use of a medication, it is often necessary to switch to a different class of medications to regain the function that has been lost or altered. Additionally, clinicians can start a medication, such as a phosphodiesterase-5 (PDE-5) inhibitor, to counteract the sexual side effects associated with the initial medication. Table 23.2 presents an overview of the sexual dysfunctions associated with various classes of medications commonly encountered in a psychiatry practice.^{25,172,255} Medications that most commonly cause sexual side effects are listed in boldface in Table 23.2. It should be noted that most research in this area has been focused on male sexual dysfunction; therefore less is known about the effects of these medications on female sexual dysfunction.

Evaluation of Sexual Dysfunction

Sexual History Taking

The idea of obtaining a sexual history can be daunting to some physicians. Remember that patients' quality of life can be significantly improved if their physicians are not afraid to discuss this topic with them. Because sexuality is a sensitive topic, it is best to maintain an attitude of openness and flexibility throughout the interview. There are multiple validated tools for assessing sexual dysfunction.¹⁴⁴ The models ALLOW, PLISSIT, and BETTER are three different approaches for facilitating such a conversation (Box 23.2).^{87,107,115}

When asking a patient about sexual dysfunction, it is important to assess the nature of the problem and the time course of the complaint (including whether it started in relation to a certain disability, medical comorbidity, or medication administration). It is equally vital to ascertain whether the patient is experiencing dissatisfaction or disruption of quality of life because of the sexual dysfunction. The Brief Sexual Symptom Checklist is a self-report tool that can be a useful adjunct to the physician's comprehensive sexual history (Fig. 23.4).¹⁰⁷

A thorough inquiry into the patient's sexual dysfunction also includes obtaining information about the medical, sexual, and psychosocial history of the patient.^{107,128} The basic medical history includes both medical and surgical history, medication use (including over-the-counter and herbal medications), substance use (including smoking, alcohol abuse, and recreational drug use), and family medical history.⁸⁷ Important medical conditions to inquire about include cardiovascular disease, diabetes, hypertension, hyperlipidemia, cancer, benign prostatic hypertrophy, or the presence of lower urinary tract symptoms, neurologic diseases (e.g., SCI, MS, TBI, and stroke), thyroid conditions, and endocrine deficiencies (e.g., hypogonadism, androgen insufficiency, or estrogen deficiency).^{28,91,107,140,224} In females, it is also necessary to ask about reproductive history, history of gynecologic disease (e.g., fibroids or endometriosis), and current menstrual status.¹⁴⁰

TABLE 23.2 Medications and Sexual Dysfunction^a

Drug Category	Drug Class ^b	Impact on Sexual Function (Most Common Listed First)
Cardiac	Diuretics (thiazides , spironolactone , loop diuretics, chlorthalidone)	ED, decreased sexual desire, impaired ejaculation, retrograde ejaculation
	Centrally acting sympatholytics (clonidine, alpha-methyldopa)	ED, decreased sexual desire
	Beta blockers	ED, decreased sexual desire (males and females)
	Alpha blockers (prazosin, terazosin)	ED, priapism (rare), retrograde ejaculation
	Vasodilators (hydralazine)	Priapism (rare)
	Antiarrhythmics (digoxin , disopyramide)	ED, decreased sexual desire
	Anticholesterolemic (statins , fibrates , niacin)	ED, decreased sexual desire
Psychiatric	Selective serotonin reuptake inhibitors	Ejaculatory dysfunction, anorgasmia (both sexes), decreased sexual desire (both sexes), ED
	Serotonin-norepinephrine reuptake inhibitors	ED, decreased sexual desire
	Tricyclic antidepressants	Decreased sexual desire, ED
	Trazodone	Priapism
	Antipsychotics	Decreased sexual drive (both sexes), ejaculatory/orgasmic dysfunction, ED, priapism
	Benzodiazepines	Orgasmic dysfunction (females), delayed ejaculation, decreased sexual desire
	Neurostimulants (methylphenidate, amantadine)	Hypersexual behavior
Gastrointestinal	H₂ blockers (especially cimetidine)	ED, decreased sexual desire, painful erections, gynecomastia
	Proton pump inhibitors	ED, gynecomastia
	Metoclopramide	Decreased sexual drive, ED
Other	Baclofen (especially intrathecal baclofen)	Ejaculatory dysfunction, ED, decreased orgasmic function (both sexes)
	Gabapentin, pregabalin, and other anticonvulsants (phenytoin)	Ejaculatory dysfunction, anorgasmia (both sexes), and decreased sexual desire (both sexes)
	Opioids	Decreased sexual desire, anorgasmia, ED, hypogonadism
	Tramadol	Delayed ejaculation
	NSAIDs	ED
	Corticosteroids	Decreased sexual desire
	Methotrexate	ED, decreased sexual desire, gynecomastia

ED, Erectile disorder/dysfunction; NSAIDs, nonsteroidal antiinflammatory drugs.

^aFrom references 25, 32, 59, 66, 104, 166, 172, 217, 231, 248, 255, 260, 266.

^bThe medications that cause sexual side effects most commonly are listed in boldface.

• BOX 23.2 ALLOW, PLISSIT, and BETTER Models for Facilitating Communication About Sexuality

A—Ask the patient about sexual function and activity.

L—Legitimize the patient's problem and acknowledge that sexual dysfunction is a relevant clinical issue.

L—Limitations in the evaluation of the sexual dysfunction should be identified.

O—Open up the discussion: Potentially refer the patient to a subspecialist.

W—Work together to develop goals and a treatment plan.

P—Permission should be obtained from the patient to discuss sexuality.

LI—Limited information should be given (e.g., about normal sexual functioning and effects of disability).

SS—Specific suggestions should be given about the patient's particular complaint.

IT—Intensive therapy may be required, including referral to a subspecialist or therapist.

B—Bring up the topic of sexuality.

E—Explain that sexuality is important and that you are open to discussion of it with your patient.

T—Tell your patient about resources that you will use to assist them.

T—Time the discussion to the patient's preference.

E—Educate the patient about the side effects of treatment medications and the disability itself.

R—Record that you had a conversation about sexuality in the patient's medical record.

Data from Frank JE, Mistretta P, Will J. Diagnosis and treatment of female sexual dysfunction. *Am Fam Physician*. 2008;77:635-645; Hatzichristou D, Rosen RC, Broderick G, et al. Clinical evaluation and management strategy for sexual dysfunction in men and women. *J Sex Med*. 2004;1:49-57; Hordern A. Intimacy and sexuality after cancer: a critical review of the literature. *Cancer Nurs*. 2008;31:e9-e17.

Brief sexual symptom checklist: men's version

Please answer the following questions about your overall sexual function in the past **3 months** or more.

1. Are you satisfied with your sexual function?

☐ Yes ☐ No

If no, please continue.

2. How long have you been dissatisfied with your sexual function?

3a. The problem(s) with your sexual function is: (mark one or more)

- ☐ 1 Problems with little or no interest in sex
- ☐ 2 Problems with erection
- ☐ 3 Problems with ejaculating too early during sexual activity
- ☐ 4 Problems taking too long, or not being able to ejaculate or have orgasm
- ☐ 5 Problems with pain during sex
- ☐ 6 Problems with penile curvature during erection
- ☐ 7 Other:

3b. Which problem is most bothersome (circle) 1 2 3 4 5 6 7

4. Would you like to talk about it with your doctor?

☐ Yes ☐ No

Brief sexual symptom checklist: women's version

Please answer the following questions about your overall sexual function in the past **3 months** or more.

1. Are you satisfied with your sexual function?

☐ Yes ☐ No

If no, please continue.

2. How long have you been dissatisfied with your sexual function?

3a. The problem(s) with your sexual function is: (mark one or more)

- ☐ 1 Problems with little or no interest in sex
- ☐ 2 Problems with decreased genital sensation (feeling)
- ☐ 3 Problems with decreased vaginal lubrication (dryness)
- ☐ 4 Problems reaching orgasm
- ☐ 5 Problems with pain during sex
- ☐ 6 Other:

3b. Which problem is most bothersome (circle) 1 2 3 4 5 6

4. Would you like to talk about it with your doctor?

☐ Yes ☐ No

• **Fig. 23.4** The Brief Sexual Symptom Checklist for males and females. (Redrawn from Hatzichristou D, Rosen RC, Broderick G, et al. Clinical evaluation and management strategy for sexual dysfunction in men and women. *J Sex Med.* 2004;1:49-57.)

The basic sexual history includes age of first sexual experience, types of sexual practices, sex of partners, history of sexually transmitted diseases, inquiry into safe sex practices, and type of birth control used.¹⁸ It should never be assumed that the patient is heterosexual, and sex-neutral terms (such as “partner” or “spouse” instead of “boyfriend” or “wife”) are preferable until the relationship has been clarified by the patient.^{16,128}

The psychosocial history relevant to sexual dysfunction includes a history of depression, anxiety, disordered sleep, or other psychiatric illness.²⁸ Major life stressors and relationship dynamics should also be discussed.^{107,140} Any history of abuse (physical, sexual, verbal, or emotional), sexual trauma, or domestic violence should also be sensitively explored.¹²⁸

Physical Examination

Although physical examination findings can be normal in patients with sexual dysfunction, it is important to do a thorough evaluation to identify pathology and to provide patient education about what is normal for the disability or disease state.⁸⁷ A comprehensive examination includes height, weight, and vital signs; auscultation of

the heart and lungs; examination of the thyroid, lymph nodes, breasts, and abdomen; a check of peripheral pulses; evaluation for lower extremity edema; a neurologic examination; and a thorough genital examination.^{224,271} In particular, it is important to look for signs of cardiovascular disease (obesity, hypertension, diminished peripheral pulses, and lower extremity edema) and endocrinopathies (thyromegaly and gynecomastia).²⁸

The neurologic examination should include mental status, motor and sensory testing; measurement of the range of motion (ROM) of joints; and evaluation of muscle tone, coordination, and reflexes.²⁷¹ ROM measurements of the hips, knees, shoulders, and hands are particularly important. It is necessary to note whether the patient has UMN or LMN findings. Sensory testing should include light touch, pinprick, and proprioception in the lower limbs.²⁷¹ Paying particular attention to the sensory evaluation of T11 to L2 and S2 to S5 is especially important because they represent the sympathetic and parasympathetic outflow tracts.^{239,271} Reflexes should be checked in the upper and lower limbs. The anal wink and bulbocavernosus reflexes both evaluate the integrity of the pudendal nerve and should be performed in both sexes.^{224,271}

In males, the genital examination consists of examining the penis for lesions and urethral position.²²⁴ The penis should also be palpated in the stretched position to detect fibrous plaques consistent with Peyronie disease (an acquired disorder of the tunica albuginea characterized by the formation of fibrous tissue and often accompanied by penile pain and deformity on erection).²⁸ The testes should be checked for size, masses, and position, and the rectum should be evaluated for sphincter tone, pelvic floor muscle tenderness and strength, prostate size, masses, and lesions.^{224,271}

In the genital examination of females, it is important to check the appearance of external genitalia, looking for inflammation or atrophy of the vulva, episiotomy or childbirth laceration scarring or strictures, and external dermatologic lesions.^{128,140} A vaginal examination with one or two fingers inserted into the vagina is a good way to assess many of the common problems associated with female sexual dysfunction.²⁷¹ Vaginismus can be appreciated by resistance or inability to insert a finger into the vagina.¹²⁸ Tenderness of the levator ani and obturator internus muscles can be assessed, as well as pelvic floor muscle strength, coordination, and hypotonicity or hypertonicity.^{87,271} The presence of pelvic organ prolapse should be noted.¹³² Bimanual examination can be performed to assess for uterine and adnexal abnormalities, including tenderness and masses.^{128,140} Rectal examination should also be performed to assess for anal sphincter muscle tone, pelvic floor muscle tenderness, and voluntary and involuntary pelvic floor muscle contraction and relaxation ability.²⁷¹

In both genders, the physician should check for signs of infection or sexually transmitted diseases, including discharge, rashes, or ulcerations.¹²⁸

Diagnostic Evaluation

Many laboratory tests and diagnostic procedures can be used to determine the pathologic origin of a patient's sexual dysfunction. In general, the selection of treatment options is based on the presenting complaint and not on the etiology behind the sexual dysfunction. Consequently, complicated diagnostic testing is usually unnecessary. Recommended laboratory testing for all males and females with sexual dysfunction includes complete blood cell count, chemistry panel, fasting glucose, and fasting lipid profile.^{107,128,178} Other laboratory testing can be warranted on the basis of history and physical examination findings, including thyroid studies and serum-free testosterone, prolactin, and prostate-specific antigen levels.^{7,28,107,180,224} Measurement of other sex hormones, such as estrogen, follicle-stimulating hormone, luteinizing hormone, or total testosterone, has been shown to have far less utility in a majority of cases.⁷ In females, androgen levels do not correlate with sexual dysfunction and are not currently recommended for diagnosis.^{69,142} Vaginal wet mount testing or screening for gonorrhea, chlamydia, or HIV can be done if infection is clinically suspected.¹²⁸

For males with ED, a variety of specialized diagnostic procedures are used to determine the specific etiology of disease, usually in preparation for surgical treatment.¹⁰⁷ The use of these tests has declined since the advent of oral PDE-5 inhibitor medications caused by their ease of use, high level of effectiveness, and low side effect profile. The procedures include penile color duplex ultrasound, nocturnal penile tumescence monitoring, pharmacocardiography, pharmacocavernosometry (PHCAS) or pharmacocavernosography (PHCAG), and electrodiagnostic testing.⁹⁵ Penile color duplex ultrasound is the most practical and commonly used diagnostic modality. It is a good tool to diagnose vasculo-

genic ED and is minimally invasive.⁹⁵ Nocturnal penile tumescence monitoring measures sleep-related erections and has traditionally been used to distinguish psychogenic and organic ED.^{95,107} Penile pharmacocardiography, PHCAS or PHCAG, and electrodiagnostic testing, such as dorsal nerve stimulation or somatosensory evoked potentials, are more invasive and time consuming, so they are rarely used.^{95,242}

In females, pelvic ultrasound is indicated if uterine or adnexal pathologic disorders are suspected. The use of objective measures for genital blood flow, such as vaginal photoplethysmography, and advanced imaging techniques, such as functional magnetic resonance imaging, are currently limited to the research setting.^{7,149}

Treatment of Sexual Dysfunction

Male Hypoactive Sexual Desire Disorder

Limited research has been done on hypoactive sexual desire disorder in males, and thus little is known about treatment options. Secondary hypoactive sexual desire in males is thought to develop most frequently in response to another type of sexual dysfunction, usually ED or premature ejaculation.¹⁶⁵ It follows that treatment of the primary sexual dysfunction would improve desire. Secondary hypoactive sexual desire disorder is also often related to medication side effects, and thus changing the drug dose or category will often improve the patient's sexual drive. Primary hypoactive sexual desire is thought to typically relate to a "sexual secret," such as a variant arousal pattern (e.g., a male who is aroused by internet pornography but not by his partner), a preference for masturbatory sex over partner sex, a history of poorly processed sexual trauma, or conflict about sexual orientation.¹⁶⁵ Treating these psychosocial factors and getting patients to be honest with themselves and their partner about their true motivations will be helpful.

Hypogonadism and low testosterone levels certainly contribute to hypoactive sexual desire in males, and treatment with testosterone supplementation is beneficial.¹⁷¹ It is less clear whether treatment with testosterone in eugonadal males with hypoactive sexual desire disorder can improve their sexual drive, but at least two studies have shown this to be the case.^{14,188}

Erectile Dysfunction

The treatment of ED was revolutionized in 1998 with the US Food and Drug Administration (FDA) approval of sildenafil, the first of the now ubiquitous PDE-5 inhibitors, which have low side effect profiles and excellent efficacy across the spectrum of disease states that cause ED.^{156,181} There are four PDE-5 inhibitors currently approved for use in the United States: sildenafil (Viagra), vardenafil (Levitra), tadalafil (Cialis), and avanafil (Stendra). Avanafil received FDA approval in 2012 and may prove to have the earliest onset of action of available PDE-5 inhibitors.¹⁹⁶ Other new PDE-5 inhibitors, such as lodenafil, udenafil, and mirodenafil, have shown promise in meta-analyses of recent clinical trials and are currently available for use in a few countries.^{75,78,108,196}

Sildenafil and vardenafil have similar pharmacokinetics, with onset of action within 30 to 120 minutes and duration of efficacy of 4 to 5 hours.⁸⁰ Tadalafil has an onset of action of 30 to 60 minutes and duration of efficacy of 12 to 36 hours.⁸⁰ High-fat intake is known to impede the action of sildenafil and vardenafil significantly.¹⁹⁶ Tadalafil has been approved for on-demand or daily dosing and is effective for the concomitant treatment of symptoms related to ED and benign prostatic hyperplasia.¹⁹⁶

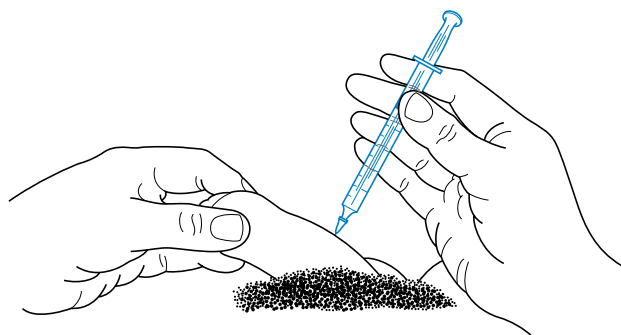
PDE-5 inhibitors have been studied and proven effective in patients with cardiovascular disease, hypertension, diabetes, SCI, MS, and depression.^{80,196} These medications have been reported to improve erectile function significantly as evidenced by successful vaginal penetration in as many as 79% to 87% of patients.¹⁹⁶ Males with SCI achieve up to 80% success rates.^{70,98} Side effects reported with PDE-5 inhibitors include headache, flushing, rhinitis, back pain, and hearing loss.⁸⁰ Nonarteritic anterior ischemic optic neuropathy has been reported, but to date, no causal link has been confirmed to PDE-5 inhibitor use.¹⁹⁶ These medications are strictly contraindicated in patients taking nitrates for chest pain.¹⁵⁶ In the setting of documented underlying hypogonadism, concurrent treatment with testosterone replacement therapy increases the effectiveness of PDE-5 inhibitors.^{178,196}

PDE-5 inhibitors are known to be less effective in patients with very severe ED, uncontrolled diabetes with neuropathy, severe vascular disease, and in those who have undergone radical prostatectomy. Currently, few other oral agents show treatment benefit. Those available include sublingual apomorphine and yohimbine.¹⁵⁶ Bremelanotide, a melanocortin receptor agonist, was efficacious in multiple clinical trials, including those in patients who had been unsuccessfully treated with sildenafil.^{215,227} Future oral therapies potentially include other melanocortin receptor agonists, dopamine receptor agonists, L-arginine, and Rho-kinase inhibitors.^{108,196} Bupropion, given in a 6-week course, has been shown to be effective in treating both hypoactive sexual desire and ED in methadone-dependent male individuals with methadone emergent sexual dysfunction.²⁷²

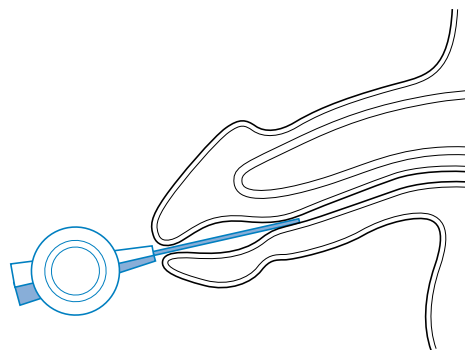
Second-line treatments after PDE-5 inhibitor treatment failure include intracavernosal injection therapy, intraurethral alprostadil (medicated urethral system for erection [MUSE]) therapy, topical alprostadil, low-intensity extracorporeal shock wave therapy (LI-ESWT), and vacuum constriction devices.

LI-ESWT is thought to address ED by inducing neovascularization and increasing blood flow in cavernosal tissues.²⁵⁸ It has been shown in several studies to be a safe and effective treatment modality for mild to moderate ED.^{131,272} However, it has not yet been approved by the FDA, and more studies are needed.

Intracavernosal penile injections have been used for decades with treatment satisfaction rates of 87% to 93.5%, relatively low adverse effects, and rapid onset of action (Fig. 23.5).^{80,156} However, considerable training is required for effective use. High discontinuation rates have been noted secondary to penile pain, and complications that can occur include priapism and Peyronie disease.⁸⁰ The most commonly injected medications include various combinations of FDA-approved alprostadil and the off-label use of papaverine and phentolamine.¹⁵⁶



• Fig. 23.5 Penile injection.



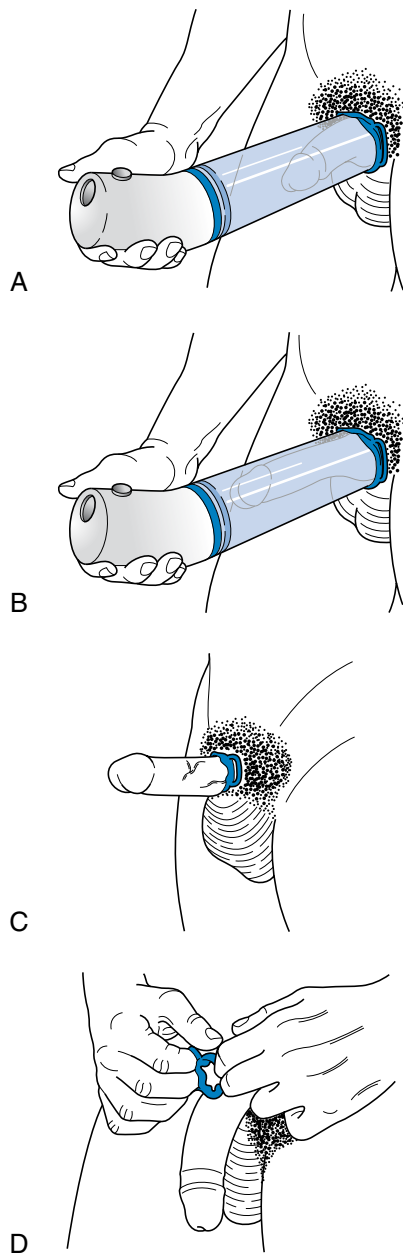
• Fig. 23.6 Intraurethral therapy.

Intraurethral treatment of alprostadil with the MUSE system can be an alternative in patients who cannot take PDE-5 inhibitors and who do not want to try more intracavernosal injection therapy (Fig. 23.6). Its efficacy ranges from 37% to 53%, and penile pain is a common complaint.¹⁹⁶ It is rarely associated with hypotension and syncope.^{80,156} Topical alprostadil formulations have undergone clinical trials with modest efficacy results, but this treatment is not currently recommended as monotherapy for ED.¹⁹⁶ Side effects include penile burning and partner vaginal pain.¹⁰⁸ Studies have shown success with PDE-5 inhibitor and alprostadil (intracavernosal, intraurethral, or topical) combination therapy for patients who have previously failed either treatment as a monotherapy.¹⁷⁶

Vacuum constriction devices apply a negative pressure to draw blood into the corpora cavernosa, which is then retained by the application of a constriction band at the base of the penis (Fig. 23.7).¹⁵⁶ Efficacy rates are as high as 90%, but satisfaction rates tend to be lower because of the unnatural appearance of the erection, penile pain, and trapped ejaculate.^{80,196} Constriction devices are contraindicated in patients with a history of severe bleeding disorder, priapism, or severe penile curvature.¹⁹⁶ Anticoagulant therapy is a relative contraindication.¹⁵⁶ These devices are often preferred in males who do not want to take medications. The constriction band cannot be left in place longer than 30 minutes because of the risk of ischemia, especially in patients who are insensate, such as those with SCI.^{80,196}

Third-line treatments for ED include surgical options, such as penile prostheses (Fig. 23.8). Penile prostheses come in two types: semirigid (malleable) and inflatable. Semirigid prostheses have malleable silicone elastomer rods with central metal cables that can be bent or straightened to produce erection.⁸⁰ Inflatable prostheses have cylinders that are implanted into the corpora cavernosa, a reservoir with fluid, and a pump that is placed in the scrotum.⁸⁰ Erection is achieved by compressing the scrotal pump to transfer fluid from the reservoir to the cylinders.⁸⁰ Infection rates have been reduced to approximately 1% because of antibiotic and hydrophilic coatings that are now available on the prostheses.⁸⁰ Mechanical failure rates are 5% at 1 year, 20% by 5 years, and 50% at 10 years.¹⁵⁶

Emerging treatments for ED continue to diversify, focusing on both central and peripheral mechanisms as well as innovative physical therapies. Central neurochemical peptides, such as melanotan II (a synthetic analogue of the peptide hormone alpha-melanocyte-stimulating hormone) and bremelanotide, demonstrate potential by triggering erections and enhancing sexual desire without the need for sexual stimulation, although they come with side effects such as nausea.¹³⁷ On the peripheral front, advancements include



• **Fig. 23.7** Use of a vacuum-assisted device. Patient instructions are as follows: (A) Place your penis inside the cylinder and use the ErecAid system's specially designed pump to produce the vacuum that pulls blood into the penis. (B) The vacuum allows you to create a near-natural erection within a few minutes. (C) After you slip the tension ring off the cylinder and onto the base of your penis and remove the cylinder, you can safely and easily keep the erection for up to 30 minutes, using only the tension ring. (D) Once you remove the tension ring, your penis returns to its soft state.

novel PDE-5 inhibitors, such as SLx-2101, which offer different administration methods and improved pharmacokinetics.¹³⁷ Additionally, therapies such as stem cell (SC) treatments are being explored, with clinical trials indicating improved penile function in ED patients through various forms of SC administration. Moreover, physical modalities such as low-intensity shock wave therapy and low-intensity pulsed ultrasound are gaining attention for their noninvasive nature and promising results in improving penile hemodynamics and tissue remodeling.¹³⁷ These emerging therapies

aim to enhance efficacy, reduce side effects, and provide more convenient and diverse treatment options for managing ED, addressing gaps left by existing treatments.

Premature Ejaculation

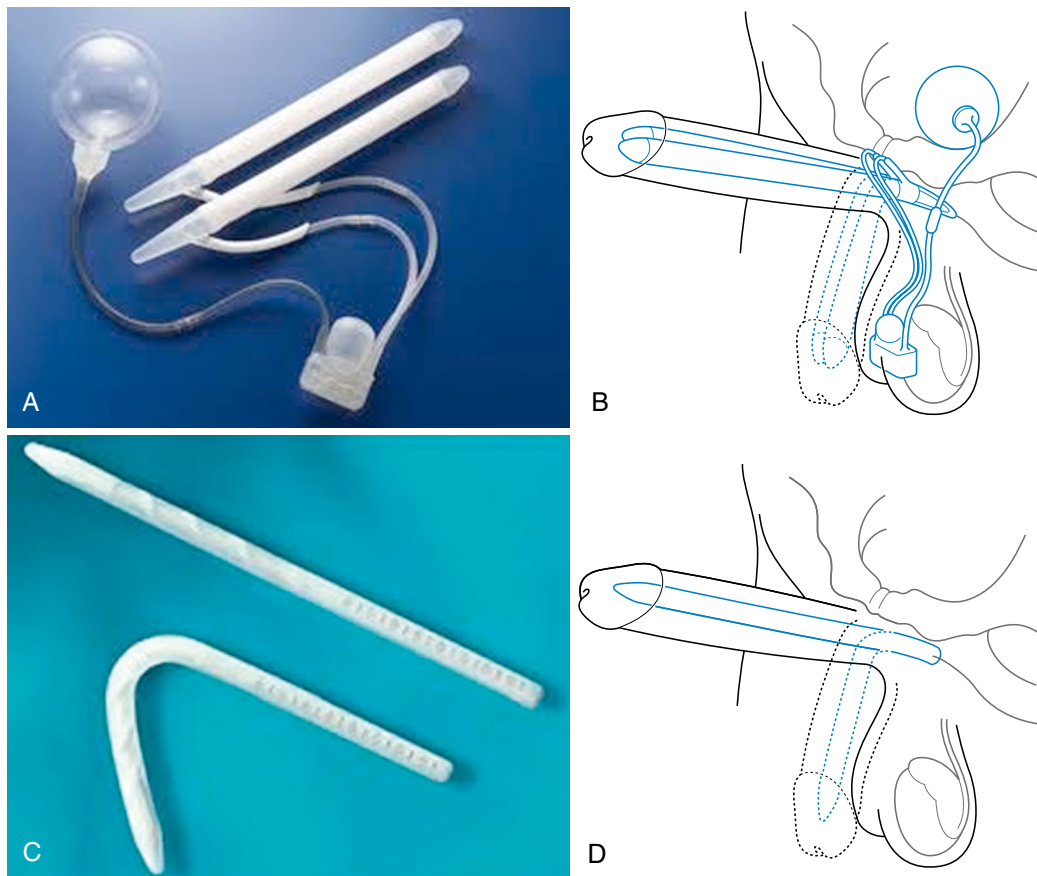
The mainstay of treatment until recently for premature ejaculation was cognitive behavioral therapy (CBT) and psychological counseling. Various CBT approaches include the start-pause and frenulum squeeze techniques, as well as experimentation with positioning, rhythm, speed, breathing, and depth of penile penetration.^{6,190,210} These strategies have success rates of up to 70% in the short term, but long-term treatment satisfaction has only been reported at 25% to 60%.²¹⁰ Cognitive behavioral approaches to treatment are often very time consuming, expensive, and perceived as intrusive and mechanistic, affecting intimacy and spontaneity during a sexual encounter.^{190,210}

Although no medications are FDA approved for treating premature ejaculation, many pharmacologic approaches have been well studied and show significant benefit. (Note that the following medications are used off-label for this condition.) Selective serotonin reuptake inhibitors (SSRIs) are the most commonly used, with paroxetine being the most effective, followed by fluoxetine and sertraline.^{6,212} The tricyclic antidepressant clomipramine has also been used successfully, but anticholinergic side effects limit its tolerance.^{6,190} Both SSRIs and clomipramine are usually given daily because of their slow onset of action (5 hours), long half-lives, and long treatment time (up to 4 weeks) to achieve a steady state.^{6,190,212}

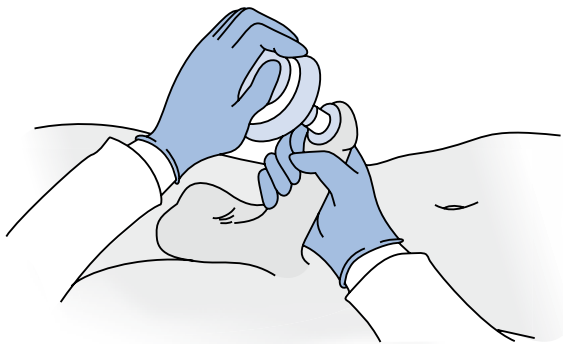
On-demand treatment for premature ejaculation has been the holy grail of research into this disorder, and some recent advances have been promising. Conventional SSRIs have been studied for as-needed use, with less profound effects on IELT vs. daily dosing.^{6,212} Dapoxetine is a short-acting SSRI developed specifically for on-demand use for premature ejaculation. It achieves peak plasma concentration within 1 hour and has a half-life of 1.5 hours.¹⁹⁰ It has been shown to significantly increase IELT and improve patient satisfaction.^{133,190} Dapoxetine has been approved for use in several countries but is not yet available in the United States.⁷ Tramadol has also been shown to be an excellent on-demand treatment for premature ejaculation, with one study showing a 13-fold increase in mean time to ejaculation.^{6,212,214,269} Longer term safety studies and comparison to SSRIs are needed. Topical agents, such as lidocaine or EMLA cream, can be helpful by decreasing sensory perception in the penis, thereby prolonging ejaculation.^{6,212} In the absence of concurrent ED, PDE-5 inhibitors are contraindicated in premature ejaculation treatment.⁶

Delayed Ejaculation, Anejaculation, and Anorgasmia in Males

Assisted ejaculation methods that have proved effective for fertility treatment in males with SCI, MS, and other disabilities include penile vibratory stimulation (PVS) and rectal probe electroejaculation (EEJ) (Figs. 23.9 and 23.10).²⁴⁶ PVS is the most commonly used technique because it is able to produce superior sperm quality, is more comfortable and preferable to patients, and can be used in a home setting.^{42,62,203} PVS produces ejaculation in only 60% to 80% of cases, whereas EEJ is 80% to 100% successful.²⁰³ EEJ is usually only used in patients in whom PVS has been unsuccessful because it produces lower quality semen, must be conducted in a physician's office setting, and often requires the use of anesthesia to perform the procedure and retrieve ejaculate.^{42,62,203}

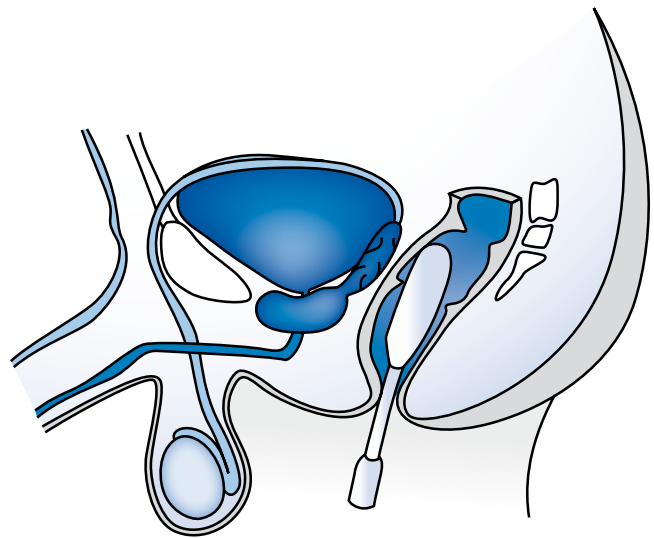


• **Fig. 23.8** Penile prostheses. (A, B) Inflatable. (C, D) Malleable.



• **Fig. 23.9** Vibrator disc position during penile vibratory stimulation. (Redrawn from Sonksen J, Ohi DA. Penile vibratory stimulation and electroejaculation in the treatment of ejaculatory dysfunction. *Int J Androl.* 2002; 25:324.)

Chemically assisted ejaculation is also a possibility, particularly with the use of midodrine to improve ejaculation success rates in combination with PVS.⁶² Vardenafil has been shown to improve ejaculation rates in males with SCI.⁹⁸ Care must be taken to monitor blood pressure and look for signs of autonomic dysreflexia in males with SCI undergoing assisted ejaculation.²⁴⁶ Deep brain stimulation of the nucleus accumbens has been proposed to stimulate orgasm for patients with SCI, but this is still in the theoretic stages.²⁵³



• **Fig. 23.10** Rectal probe position during an electroejaculation procedure. (Redrawn from Sonksen J, Ohi DA. Penile vibratory stimulation and electroejaculation in the treatment of ejaculatory dysfunction. *Int J Androl.* 2002;25:324.)

Delayed ejaculation in males is much harder to treat when the goal is enhanced sexual function as opposed to fertility. Pharmacologic treatment for ejaculatory and orgasmic dysfunction in males is still largely unproved, although there have been some case series and small controlled trials with sildenafil, bupropion, amantadine,

bupirone, cyproheptadine, and yohimbine that showed modest benefits.^{168,169,175} Cognitive behavioral modification and psychotherapy are indicated in males with psychogenic (situational or relational) orgasmic dysfunction.^{1,156}

Female Sexual Interest/Arousal Disorder

When considering treatment for females with low sexual desire or arousal, it is important to remember that the sexual response cycle of females is very different from that of males. Many females experience only responsive, as opposed to spontaneous, sexual desire.¹⁹ The first step in treating a female with a complaint of low sexual desire is educating the patient as to what is normal for females because there are many misconceptions fueled by popular culture about what females should feel.⁸⁷ Referring the patient to the work of Basson may facilitate reassurance and improved understanding.¹⁷⁻¹⁹ After education, the next intervention level includes modification of potentially contributing factors such as lack of lubrication, menopausal vasomotor symptoms, pelvic pain, sleep dysfunction, or depression.⁵⁸

Despite limited research, psychotherapy remains a mainstay for the treatment of female sexual interest/arousal disorder. In one of the few empirically tested outcome studies, mindfulness-based CBT improved sexual desire and arousal in gynecologic cancer survivors.⁴⁷ Sexual behavioral techniques, sex therapy, and couples counseling (originally conceived by Masters and Johnson)¹⁶⁰ have been shown to be beneficial by helping to reduce anxiety and exaggerated sexual expectations.^{160,164}

Flibanserin oral tablets and bremelanotide injectables are currently the only FDA-approved medications for treating generalized acquired hypoactive sexual desire disorder in premenopausal females. Flibanserin, categorized as a multifunctional serotonin agonist and antagonist, typically elicits a response in about 50% of treated females, with efficacy potentially taking up to 8 weeks to manifest.^{73,132,256} Although clinical trials in postmenopausal females have shown comparable effectiveness, flibanserin is not currently approved by the FDA for this demographic.²³² Conversely, the bremelanotide injection operates by activating melanocortin receptors, albeit its precise mechanism remains unknown. Clinical trials demonstrated that patients using injectable bremelanotide experienced increased sexual desire scores and decreased distress scores compared with placebo, although common side effects such as nausea and vomiting were reported. However, caution is advised for individuals with high blood pressure or cardiovascular disease, and it should not be used concomitantly with certain medications, such as naltrexone.¹⁶³

The other major pharmacologic treatment for female hypoactive sexual desire is testosterone therapy. Many treatment preparations are available, but a transdermal patch is the most commonly studied and prescribed method.¹⁹² There have been studies that have demonstrated the effectiveness of transdermal testosterone for increasing sexual desire and satisfaction in postmenopausal females or surgically menopausal females who have had bilateral oophorectomies.^{43,84,230} Less robust data exist in premenopausal females, and potential fetal exposure remains a concern.^{67,142} A recent systematic review failed to show correlation between serum testosterone levels and libido in premenopausal females.¹⁹⁹ Side effects are usually related to masculinization, such as hirsutism and acne. The FDA has not approved the use of testosterone to treat female sexual interest/arousal disorder, and long-term safety data have not been established.⁸⁴ Theoretic concerns include effects on breast and endometrial cancer risks and cardiovascular

health. Current available safety data, although not conclusive, are reassuring with regard to breast, endometrial, and cardiovascular outcomes.^{68,142} Further research is necessary.

Other medications that show promise for treating female hypoactive sexual desire include tibolone (a synthetic steroid), melanocortins, and combinations of testosterone with PDE-5 inhibitors or bupirone.^{74,142,164,227}

Pharmacologic interventions for female arousal and lubrication disorders have been increasingly studied in recent years, yielding mixed results. Because the physiology of female arousal is thought to mirror male erectile function, there was significant hope that sildenafil would be an effective treatment. However, research trials have been largely disappointing, especially compared with the success of PDE-5 inhibitors in males with ED. Although sildenafil has been well tolerated, the overall body of research to date indicates it does not benefit most females with generalized arousal disorder without a known organic cause.^{24,164,229} For females with disabilities, PDE-5 inhibitors remain a viable treatment option. Trials have shown positive results in small populations of females with acquired genital arousal disorders of an organic nature, such as premenopausal type 1 diabetes, SCI, MS, and those taking SSRIs.^{53,164,187,241} More research is needed on PDE-5 inhibitors in females, both with and without acquired disabilities.

Other medications have shown more promise than sildenafil for treating female arousal disorders, especially in postmenopausal females. These include melanocortins, phentolamine, l-arginine, and alprostadil. Acquired arousal disorder in postmenopausal females is often treated with systemic or local estrogen therapy to improve vaginal lubrication and blood flow.¹⁴¹ Mechanical devices can also enhance arousal, such as vibratory stimulators and the Eros Clitoral Therapy Device—a clitoral vacuum approved by the FDA to boost female arousal and orgasm.^{74,119,135,211,213,227}

Female Orgasmic Dysfunction

Female orgasmic dysfunction, when not accompanied by an arousal disorder, is typically considered a psychogenic issue. Effective treatments include CBT, which aims to reduce anxiety and shift sexual attitudes and thoughts; sensate focus therapy, which involves guided exercises to enhance sexual intimacy with a partner; and directed masturbation, which educates females on the normalcy and health benefits of self-stimulation to discover what works for them.^{87,141} These behavioral treatments have been shown to be effective in 60% of females.¹⁷⁰ Bupropion has been shown in at least one study to help orgasmic dysfunction in females.¹⁷⁵ As with decreased arousal, mechanical devices can help with the attainment of orgasm.³³

Genitopelvic Pain/Penetration Disorder

Dyspareunia and vaginismus have various organic and psychogenic causes, and proper treatment depends on identifying the underlying etiology. Diagnosing and treating any underlying pelvic pathology is crucial. The management of pelvic floor dysfunction and vulvodynia, major contributors to genitopelvic pain/penetration disorder, is detailed in Chapter 38.

Summary

Disability does not lessen the significance of sexuality in an individual's overall quality of life. By thoroughly understanding the

diagnosis and treatment of sexual dysfunction within the context of disability and being open to discussions about sexuality with patients, physicians can significantly enhance quality of life, providing comfort and hope to those who need it most.

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